

Supplementary Methods

Table 1. Software versions, execution environments and roles of the main components used in the somatic variant analysis

Software or component	Executed version or environment	Role in the analysis
Nextflow	26.04.4, build 12445	Workflow orchestration, process dependency management and parallel execution
Apptainer	1.4.1	Execution of containerised bioinformatic tools
OpenJDK	17.0.18-internal	Java runtime supporting GATK, VarScan2 and VarDict
GATK Mutect2	4.6.2.0	Tumour-only somatic SNV and small-indel candidate generation
GATK FilterMutectCalls	4.7.0.0	Assignment of Mutect2 caller-level filtering labels
samtools	1.23.1	Target-restricted pileup generation for VarScan2 and BAM processing
VarScan2	2.4.6	Pileup-based SNV and indel candidate generation
DeepSomatic / DeepVariant	1.10.0	Deep-learning-based tumour-only somatic variant candidate generation
VarDict	1.8.2, as reported in the executed VCF header	BAP1-focused SNV, indel and complex-variant candidate generation
Pisces	5.2.9.122	Targeted low-frequency SNV and indel candidate generation
BCFtools / HTSlib	BCFtools 1.23.1 for normalisation and 1.24 for annotation, merging and viewing; HTSlib 1.23.1	VCF decomposition, normalisation, merging, chromosome renaming, compression and indexing
Ensembl Variant Effect Predictor	VEP 115.0; API and cache release 115; GRCh38	Functional consequence and transcript annotation
DREAMS-vc environment	Custom container dreams-vc_1.0.sif	Neural-network-based evaluation of predefined uveal melanoma SNV sites
Custom read extractor	Python 3.12.13; pysam 0.23.0	CIGAR-aware extraction and summarisation of reference- and alternative-supporting reads and molecules
Versions were obtained from executed workflow records, tool environments, VCF headers or container metadata, as applicable.		

Table 2. Analysis modes and executed settings for the six somatic variant callers used in the targeted sequencing workflow.

Caller	Analysis mode and target	Caller settings
Mutect2	Tumour-only targeted calling against GRCh38	GATK Mutect2 4.6.2.0; custom panel of normals; af-only-gnomad.hg38.vcf.gz germline population resource; target interval list; F1R2 archive generation; tumour LOD to emit 3.0; initial tumour LOD 2.0; maximum population AF 0.01; minimum allele fraction 0; minimum mapping quality 20; four native PairHMM threads
VarScan2	Independent single-sample pileup calling; SNVs and indels called separately	samtools mpileup restricted to the target BED with BAQ disabled, minimum pileup base quality 0 and maximum depth 100,000; VarScan mpileup2snp and mpileup2indel; minimum coverage 500; minimum two alternative observations; minimum VAF 0.001; calling P-value 0.99; strand filter disabled; VCF output enabled
DeepSomatic	Tumour-only deep-learning calling on targeted sequencing data	DeepSomatic 1.10.0; WGS_TUMOR_ONLY model; target BED; custom panel of normals constructed from six healthy controls; minimum supporting count of 1 for SNVs and indels; minimum candidate fraction of 0.0001 for SNVs and indels
VarDict	BAP1-targeted calling using a BAP1-specific BED	Executed VarDict version 1.8.2; minimum allele fraction -f 0.0001; --fisher; column settings -c 1 -S 2 -E 3 -g 4; VCF conversion using -N [sam -E -f 0.0001
Pisces	Targeted small-variant calling on sample-specific minimal BAM files; final NL40 configuration	Pisces 5.2.9.122; -g WholeGenomeFasta; -i twist_targets.bed; -gVCF false; -CallIMNVs false; -MinVF 0.0001; -SSFilter true; -MinBQ 50; -NL -MaxVQ 100; -MinDepth 5; -MinDepthFilter 500; -MinVQ 0; -VQFilter 30; -ReportNoCalls true; -RMxNFilter 5,9,0.35; -threadbychr true
DREAMS-vc	Protocol-specific neural-network assessment of predefined uveal melanoma SNV sites rather than unrestricted de novo calling	Predefined candidate catalogue; protocol-specific HDF5 model dreams_model_HC_4.h5; candidates retained when mutation_detected = TR
Analysis modes and settings reflect the executed workflow.		

Table 3. Rule-based criteria used to retain, review or exclude candidate variants in BAP1, SF3B1 and EIF1AX.

Filter	Exact implemented criterion	Result
BAP1 recurrence filter	A BAP1 variant observed in ≥ 3 patients is excluded only when supported by exactly one caller and one alternate molecule. Other recurrent BAP1 combinations proceed to later assessment.	Exclude recurrent weak calls; otherwise retain or review.
SF3B1/EIF1AX support filter	A single-molecule SF3B1 or EIF1AX call requires median AD ≥ 5 and median BD ≥ 5 . Calls with ≥ 2 alternate molecules do not receive this additional threshold.	Pass or exclude.
BAP1 molecule-support filter	BAP1 calls with ≥ 2 alternate molecules pass. A single-molecule call passes when AD and BD are both ≥ 10 , enters review when both are ≥ 5 but either is < 10 , and otherwise fails.	Pass, review or exclude.
BAP1 functional-evidence filter	Retain technically eligible SGE-confirmed loss-of-function calls or, when no SGE match is available, frameshift, stop-gained, splice-acceptor or splice-donor calls. Recurrent LOF and technically low-support LOF calls enter review. SGE-unchanged, SGE-enriched and calls without LOF evidence are excluded.	Retain, review or exclude.